Proposal Project
Data Warehouse

S1 Data Science, Faculty of Mathematics, Science, and Natural Sciences
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Informasi Umum Proyek

Title	Analysis of the Relationship between U.S. Electricity Consumption and Global Economic Indicators: Implementation of a Data Warehouse and Business Intelligence System (2022–2024)
Topic	Economy
Revision?	No.

Description Problem formulation	Energy and economic systems are deeply intertwined, yet they are rarely analyzed together within a unified, structured analytical pipeline. The United States, as the world's largest economy and one of the top energy consumers, serves as a critical subject for understanding how electricity production and consumption patterns correlate with broader macroeconomic conditions both domestically and globally. The urgency of this problem stems from three converging trends. First, the 2022–2024 period witnessed extraordinary volatility in energy markets following post-pandemic economic recovery, the Russia-Ukraine conflict's impact on global commodity prices, and accelerating renewable energy transitions. Second, inflation surged across major economies during this period, yet the specific relationship between electricity pricing and consumer price indices has not been systematically explored through a data warehouse approach. Third, industrial production, a key driver of electricity demand, fluctuated significantly across G20 economies, creating a gap in understanding how economic output cycles correspond to energy generation mix shifts. The core challenge lies in data heterogeneity: energy data (EIA) and economic data (World Bank GEM) are published in different formats, different temporal granularities, and through different access mechanisms. Integrating them into a single analytical system that supports multidimensional OLAP queries requires a well-designed ETL pipeline and a proper dimensional data model, a challenge that existing studies rarely address end-to-end in an academic project context.
Target insight to be achieved	The crucial step in achieving our analysis insights would be the datamart itself. Through a robust data staging and data presentation pipeline through CUBE/ROLLUP/GROUPING queries on an OLAP cube. The project mainly targets these following insights: a. First, **temporal trend analysis**: identifying how U.S. net electricity generation by fuel type (coal, natural gas, nuclear, wind, solar, hydroelectric) evolved month-by-month from January 2022 to December 2024, and whether

	renewable energy generation shows a statistically consistent upward trend across all three years. b. Second, price-inflation correlation: examining whether movements in U.S. retail electricity prices (cents/kWh) across residential, commercial, and industrial sectors are correlated with CPI percentage changes in major economies (United States, Germany, Japan, United Kingdom, France, Brazil, China, India) over the same period. c. Third, industrial production and electricity demand: determining whether fluctuations in industrial production indices across target countries correspond to changes in U.S. industrial sector electricity consumption, revealing potential leading or lagging indicator relationships. d. Fourth, energy mix transition insight: using CUBE queries to drill down into quarterly aggregations of generation by fuel category (fossil vs. renewable vs. nuclear) and measure the rate of energy mix transition across the three-year observation window. e. Fifth, cross-dimensional OLAP analysis: leveraging the Star Schema to perform slice-and-dice operations across the time, fuel type, sector, and country dimensions simultaneously, enabling compound analytical questions such as "how did the renewable energy share in Q2 2022 compare to Q2 2023 in the context of global GDP growth? These insights are produced by slicing and dicing the procured cube along the time hierarchy (year → quarter → month), time, country, and fuel-type dimensions.
Research references	*https://doi.org/10.1155/2022/6884273* *https://doi.org/10.1007/s11356-022-21099-9* *https://doi.org/10.1016/j.eneco.2023.106893* *https://doi.org/10.1016/j.eneco.2023.107050* *https://doi.org/10.5547/01956574.40.6.ccon* *https://doi.org/10.12783/ijes.2015.0501.01* *https://doi.org/10.3390/en14123429*

Informasi Detail Dataset

Sumber dataset	This project utilizes two primary data sources through different mechanisms. The first source is the U.S. Energy Information Administration (EIA) Open Data API v2, accessible at https://api.eia.gov/v2/. Data is acquired programmatically using Python's request library with a registered API key (free registration at https://www.eia.gov/opendata/register.php) Two specific endpoints are used: /electricity/electric-power-operational-data/data for monthly net electricity generation by fuel type and location, and /electricity/retail-sales/data for monthly retail electricity sales, average price, revenue, and customer count by sector and state. All requests are filtered to location=us, frequency=monthly, the time range of January, 2022 - December, 2024. The second source is the World Bank Global Economic Monitor (GEM), accessible at https://datacatalog.worldbank.org/search/dataset/0037798. Data is downloaded as Excel (.xlsx) files, each containing one economic indicator across all available countries and time periods, Five indicator files are used: CPI % y-o-y nominal seasonally adjusted, GDP at market prices current US$ millions seasonally adjusted, Industrial Production constant 2010 US$ seasonally adjusted, Unemployment Rate seasonally adjusted, and Exchange Rate new LCU per USD extended backward period average.

	These two sources were chosen as they provide us with the overlapping monthly granularity from 2022 to 2024 and thus fulfilling our requirements of "*integration of heterogeneous sources."*
General description of dataset / data dump	The EIA generation dataset contains monthly records of utility-scale electricity net generation in thousand megawatt-hours, segmented by fuel type (coal, natural gas, nuclear, wind, solar, hydroelectric, geothermal, and other) and by electric power sector. The retail sales dataset contains monthly records of megawatt-hours sold, average price in cents per kilowatt-hour, revenue in millions of dollars, and number of customers, segmented by sector (residential, commercial, industrial, and all sectors) at the U.S. national level. Combined, the two EIA datasets produce approximately 800-1,200 records covering 36 months. The World Bank GEM dataset covers 8 target countries: the United States, China, Germany, Japan, India, the United Kingdom, France, and Brazil. Each indicator file contains monthly or annual observations from 1996 onward for over 100 countries but is filtered to the 8 target countries and the 2022-2024 period for this project. After filtering, each monthly indicator file yields 36 time periods x 8 countries = 288 records per indicator. GDP is an exception, available only at annual frequency, which is expanded to monthly through forward-fill methodology.
Specific aspects of the dataset	**Sparsity:** The combined dataset exhibits moderate sparsity. The Unemployment Rate indicator is the most sparse; only 6 of the 8 target countries have data available (India and China are not reported in World Bank GEM unemployment), resulting in approximately 25% sparsity for that column in the fact table. Other World Bank indicators are complete for all 8 target countries. EIA data is complete with no missing values for the US national aggregate level. **Imbalance:** The dataset is not class-imbalanced in the traditional sense, as this is a regression/analytical warehouse rather than a classification task. However, there is a distributional imbalance in generation volume across fuel types: natural gas generation is approximately 3 - 4x higher than any other single fuel type, while geothermal contributes less than 1% of total generation. This imbalance is preserved intentionally as it reflects real-world energy mix proportions. **Outliers:** The CPI dataset contains notable outliers in 2022 for several countries, reflecting the global inflation surge. The U.S. CPI peaked at approximately 9.1% y-o-y in June 2022, which is significantly above the historical mean of 2–3% for developed economies. These values are treated as genuine data points rather than errors, as they represent actual economic events. GDP values for China and the United States are orders of magnitude larger than other target countries (in absolute USD terms), which is handled through normalization during the Transform phase. **Temporal resolution mismatch:** GDP data is only available at annual granularity, while all other indicators are available monthly. This mismatch is addressed through forward-fill expansion during Transform.

Kemajuan Pengerjaan Data Staging

Extraction Step	The extraction process was fully automated using an Apache Airflow 3.0 orchestration pipeline to ensure continuous and scheduled data ingestion without manual intervention. The pipeline was configured to execute every five minutes, enabling near real-time synchronization of both energy and macroeconomic datasets into the staging environment.

	For the energy domain, monthly time-series data covering the 2022–2024 period were retrieved from the U.S. Energy Information Administration (EIA) API v2 in JSON format. The extracted datasets included U.S. retail electricity prices across multiple sectors as well as electricity generation statistics categorized by fuel type. API requests were dynamically parameterized within PythonOperator tasks in Airflow, allowing automated retrieval of updated records while maintaining consistency in temporal granularity and metadata structure. Simultaneously, macroeconomic data were extracted from five World Bank datasets provided in CSV format, consisting of Consumer Price Index (CPI), Gross Domestic Product (GDP), Industrial Production Index, unemployment rates, and exchange rates. The datasets represented eight target countries selected for comparative global economic analysis. Instead of relying on direct API requests during execution, the CSV files were preloaded into the Airflow server and accessed directly from the /raw/ directory to improve extraction stability, reduce external dependency failures, and optimize pipeline execution time. During the extraction stage, the pipeline also performed preliminary validation procedures, including file availability checks, schema consistency verification, and automated logging of extraction status. All successfully extracted raw datasets were temporarily stored in the staging layer before entering the transformation phase. This architecture ensured that heterogeneous data sources originating from both REST-based APIs and flat-file repositories could be systematically integrated into a unified analytical workflow for downstream ETL processing and OLAP analysis.
Transformation Step	The transformation phase was designed to integrate heterogeneous energy and macroeconomic datasets into a unified analytical structure suitable for OLAP and Business Intelligence processing. Since the extracted datasets originated from different sources, formats, and temporal granularities, multiple transformation procedures were required to ensure structural consistency, semantic alignment, and analytical readiness. The first transformation step involved schema normalization. All attribute names from both the EIA and World Bank datasets were standardized into snake_case format to improve readability, maintain naming consistency, and simplify downstream SQL querying and dimensional modeling. Redundant symbols, whitespace characters, and inconsistent capitalization were removed during this process. Next, temporal harmonization was performed to reconcile differences in time granularity between datasets. The electricity datasets from the EIA API were already provided in monthly format (YYYY-MM), whereas several macroeconomic indicators from the World Bank, particularly GDP, were available only in annual format (YYYY.0). To align these datasets within a monthly analytical framework, annual values were transformed into monthly observations using a forward-fill strategy, where each yearly value was propagated across all twelve months within the corresponding year. This approach enabled direct temporal comparison between energy consumption patterns and macroeconomic fluctuations. Following temporal standardization, all World Bank datasets underwent structural reshaping from wide format into long format using Pandas transformation operations such as melt(). Originally, each country and year combination appeared as separate columns, which limited analytical flexibility. The transformation produced a normalized tabular structure consisting of key fields such as country, indicator, period, and value, thereby improving compatibility with dimensional modeling and OLAP cube construction.

	Additional data cleaning procedures were then applied to improve data quality and integrity. Missing values were identified and handled selectively depending on indicator characteristics, duplicate records were removed, inconsistent country labels were standardized, and incompatible data types were converted into appropriate numeric or datetime formats. Numerical indicators were also validated to ensure proper aggregation and analytical consistency during later OLAP operations. After the normalization and cleaning processes were completed, the transformed energy and macroeconomic datasets were integrated using pd.merge() operations based on shared temporal attributes and contextual identifiers. This integration step produced a consolidated analytical dataset capable of supporting multidimensional analysis across time, sector, fuel type, and macroeconomic indicators. The final output of the transformation phase consisted of six cleaned and structured tables: two fact tables containing quantitative measurements related to electricity generation and economic indicators, and four dimension tables representing descriptive analytical contexts such as time, fuel type, sector, and country. These transformed tables were subsequently prepared for the loading stage into the PostgreSQL based data warehouse hosted on Supabase.
Load Step	The loading phase was conducted by transferring the transformed datasets into Supabase, a cloud-based PostgreSQL database, through an automated ETL pipeline orchestrated using Apache Airflow 3.0. Prior to insertion, an idempotent loading strategy was implemented using a **DELETE per period to INSERT** mechanism to ensure that repeated executions of the pipeline would not generate duplicate records. This strategy is particularly important because the pipeline is scheduled to run every five minutes for continuous data synchronization and refresh. During the loading process, two fact tables and four dimension tables were successfully populated. The **fact_energy_econom**y table stores integrated electricity and macroeconomic indicators, while the **fact_generation** table stores electricity generation metrics categorized by fuel type. In total, 144 rows were inserted into **fact_energy_economy** and 288 rows into **fact_generation**. All loading operations completed successfully without transaction failures or timeout issues, indicating that the ETL workflow is stable for periodic execution. The current database size stored in Supabase is relatively small, The current database size is compact, reflecting the focused scope of the 2022-2024 observation period and eight target countries; exact size measurement will be reported alongside performance benchmarking in the next phase. consisting of approximately 432 rows across the primary fact tables and fewer than 50 rows within the dimension tables. The compact size is primarily due to the limited observation period (2022–2024) and the restricted number of target countries analyzed. Despite the relatively small volume, the database structure has been optimized to support scalable OLAP and Business Intelligence operations. Several PostgreSQL technologies and optimization features have been implemented within the database environment. First, Table partitioning by year was defined in the schema design for the fact tables to improve query efficiency on time-range filters. to improve query efficiency and facilitate maintenance of time-series data. Second, B-tree indexing was created on frequently queried attributes such as **period** and **fuel_type_id** to accelerate filtering, aggregation, and join operations. Third, two materialized views **mv_generation_monthly** and **mv_retail_economic** were developed to pre-aggregate monthly analytical data and reduce computational overhead during OLAP queries. In addition, the **pg_stat_statements** PostgreSQL

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<tr><td></td><td>extension was enabled to support monitoring and future benchmarking of query performance.

At the current stage, quantitative performance benchmarking has not yet been fully conducted. Although the benchmarking infrastructure has been prepared through the activation of pg_stat_statements, comparative measurements such as query execution time with versus without materialized views, indexed versus non-indexed queries, and performance under repeated ETL execution have not yet been evaluated experimentally. Nevertheless, preliminary qualitative observations indicate that the pipeline is capable of completing extraction, transformation, and loading tasks within the predefined five-minute scheduling interval without operational issues. Comprehensive performance benchmarking is planned for the subsequent stage alongside OLAP implementation using Atoti DataMart.</td></tr>
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Kemajuan Pengerjaan Data Presentation

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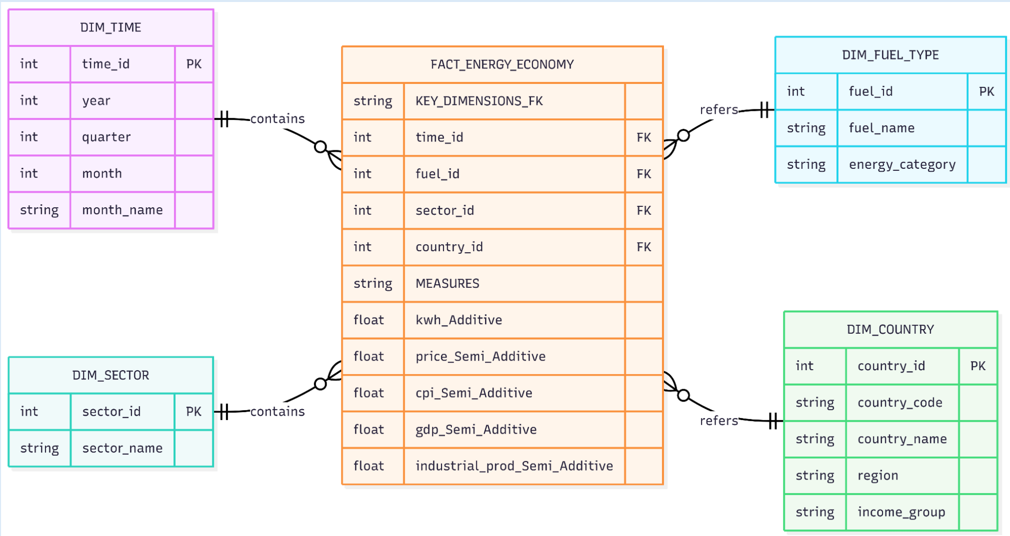
<tr><td>Subject focus</td><td>Four OLAP analysis subjects have been identified:
Dynamics of U.S. retail electricity prices by sector (2022-2024)
Correlation between electricity price movements and global macroeconomic indicators (CPI, GDP, Industrial Production)
U.S. energy mix transition from fossil fuels to renewable energy
Detection of simultaneous anomaly periods (June-August 2022) based on cross-indicator Z-scores.</td></tr>
<tr><td>Star schema</td><td>



The star schema consists of 2 fact tables and 4 dimension tables:

Fact tables:
fact_energy_economy - 144 rows; measures: retail electricity price, CPI, GDP, industrial production, unemployment, exchange rate; links energy data (EIA) with economic indicators (World Bank)
fact_generation - 288 rows; measures: electricity generation volume by fuel type; supports energy mix transition analysis
Dimension tables (OLAP axes):
dim_time (hierarchy): year → quarter → month; enables roll-up/drill-down across time
dim_sector (residential, commercial, industrial); enables slice-and-dice by consumer sector</td></tr>
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dim_country (8 target countries); enables slicing by geography for macroeconomic comparison
dim_fuel_type (fossil vs. renewable categories); enables energy mix analysis

All four dimensions form the OLAP analysis axes, enabling drill-down, roll-up, and slice-and-dice operations aligned with the research hypotheses (H1–H4).</td></tr>
<tr><td>Hasil OLAP</td><td>The complete ETL pipeline has been completed successfully, and Apache Airflow 3.0 automatically schedules it every five minutes. Two Materialised Views (mv_generation_monthly and mv_retail_economic) have been updated, and data has been put into Supabase (PostgreSQL cloud).

However, statistical testing for the four hypotheses (H1–H4) and OLAP exploration utilising the Atoti DataMart have not yet been carried out and are intended to be finished by the Final Seminar on June 2–3, 2026.</td></tr>
<tr><td>Visualisasi OLAP</td><td>Visualization has not yet been produced at this stage. Planned visualizations include:

Atoti Widget: time-series chart of retail electricity prices vs. CPI
Correlation heatmap across all indicators
Stacked bar chart of energy mix by year (fossil vs. renewable)
Scatter plot for Z-score anomaly detection

All visualizations will be implemented alongside the Atoti DataMart build in the next phase.</td></tr>
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